

YEAR 10 SCIENCE

PHYSICS

FORCES, MOTION and ENERGY

TOPIC TEST

SECTION A : MULTIPLE CHOICE	20 QUESTIONS	20m
SECTION B : WRITTEN SECTION	8 QUESTIONS	28m
	TOTAL	48m

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FORCES MOTION AND ENERGY
TOPIC TEST

MULTIPLE CHOICE

CONSTANTS: Acceleration due to gravity $g = 9.80 \text{ ms}^{-2}$

1. Which of the following is a scalar quantity?
 - (a) velocity
 - (b) acceleration
 - (c) energy
 - (d) force

2. Which one of the following masses is expressed in standard scientific notation?
 - (a) 0.0610×10^{-2}
 - (b) 3.275×10^5
 - (c) 6×10^9
 - (d) 6.37×10^3

3. An aeroplane is travelling from airport 1 to airport 2, which is due west of airport 1. For most of its flight the plane encounters a steady wind from the south. For the plane to travel along a straight line path from airport 1 to airport 2, the pilot should head the nose of the plane along a compass bearing which is:
 - (a) due west.
 - (b) due south.
 - (c) between west and north.
 - (d) between west and south.

4. A car has an initial velocity of 15.0 ms^{-1} east and uniformly accelerates to 25.0 ms^{-1} in 4.00 s. How far has the car moved during this change of velocity?
 - (a) $1.40 \times 10^2 \text{ m}$
 - (b) 50.0 m
 - (c) 80.0 m
 - (d) 65.0 m

5. An object has a mass of 0.25 tonne . It would have a weight of close to
 - (a) 25 N
 - (b) 250N
 - (c) 2500N
 - (d) none of the above.

6. Which of the following is true for an object thrown up in a vertical direction.
- (a) the total energy increases
 - (b) the total energy is constant
 - (c) the total energy always increases
 - (d) the gain in E_p is equal to the gain in E_k
7. A device for converting electrical energy into mechanical energy is
- (a) an electric oven
 - (b) a generator
 - (c) an electric motor
 - (d) a light globe.

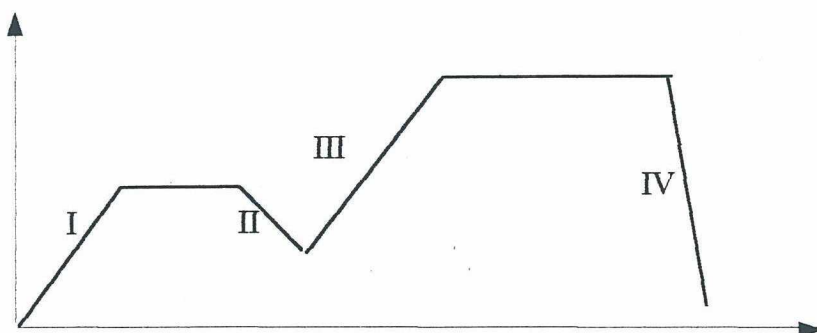
THE NEXT 3 QUESTIONS REFER TO THE FOLLOWING INFORMATION.

A body falls from rest near the Earth's surface. Ignore the effects of any air resistance.

8. The velocity of the body after 1.00 s of fall is:
- (a) 4.90 ms^{-1}
 - (b) 9.80 ms^{-1}
 - (c) 19.6 ms^{-1}
 - (d) unable to be calculated from the information.
9. The distance the body falls in the first 2.00 s is:
- (a) 4.90 m
 - (b) 9.80 m
 - (c) 19.6 m
 - (d) unable to be calculated from the information.
10. If at a certain instant the velocity of the body is 29.4 ms^{-1} , its velocity 1.00 s later is:
- (a) 29.4 ms^{-1}
 - (b) 34.3 ms^{-1}
 - (c) 39.2 ms^{-1}
 - (d) 44.1 ms^{-1}
11. A car has an initial velocity of 20.0 ms^{-1} and slows uniformly to 5.00 ms^{-1} in 4.00s. In this time it travels:
- (a) 25.0 m
 - (b) 50.0 m
 - (c) $1.00 \times 10^2 \text{ m}$
 - (d) $2.00 \times 10^2 \text{ m}$

THE NEXT 3 QUESTIONS REFER TO THE FOLLOWING INFORMATION.

The velocity-time graph below illustrates the motion of a body moving in a straight line.



12. During which of the periods I, II, III and IV, was the magnitude of the average acceleration greatest?
- (a) I
 - (b) II
 - (c) III
 - (d) IV
13. When was the object stationary?
- (a) At the start of period I.
 - (b) During period III.
 - (c) At the end of period IV.
 - (d) At none of these times.
14. During which of the periods did the object travel the greatest distance?
- (a) I
 - (b) II
 - (c) III
 - (d) IV
15. A SONAR impulse (a sound wave) is transmitted vertically downward from a stationary boat and the signal is reflected from the sea bed. It is received on the boat 10.0 s after the impulse is sent. Given that the speed of sound in the water is $1.45 \times 10^3 \text{ ms}^{-1}$, how deep is the water in this part of the ocean?
- (a) 725 m
 - (b) $1.45 \times 10^3 \text{ m}$
 - (c) $7.25 \times 10^3 \text{ m}$
 - (d) $1.45 \times 10^4 \text{ m}$

16. Which one of the following is the form of energy found in food
- (a) electrical energy
 - (b) chemical energy
 - (c) kinetic energy
 - (d) kinetic energy and chemical energy
17. Which of these is not a type of kinetic energy
- (a) translation energy
 - (b) rotational energy
 - (c) vibrational energy
 - (d) positional energy
18. Electrical energy is best described as
- (a) energy associated with motion.
 - (b) energy due to position or form
 - (c) energy associated with the movement of charge
 - (d) energy due to the forces in the nuclei of atoms.
19. The energy of an object because of its form is
- (a) magnetic energy
 - (b) potential energy
 - (c) chemical energy
 - (d) none of these
20. A pellet having 5 J of E_k is brought to rest after penetrating 0.1m into a block of clay. Assuming the clay offers uniform resistance to the pellet until it comes to rest, what is the value of this resistive force?
- (a) 0.02 N
 - (b) 0.5 N
 - (c) 20.0 N
 - (d) 50.0 N

END OF SECTION A